

Cable-based technologies

Coaxial cable

which can achieve >1Gbps or greater. Suitable for long distance, but expensive for high speed cables.

Twisted pair

Twisted pairs work on the principle of differential signalling, where equal and opposite signals are sent over the two cables in the pair and by combining them at the receiver one can mitigate noise that occurs over the channel.

cat 5 for 1Gbps or cat 6 for 10Gbps. However, short reach ~ 100m.

Copper has a faster signal propagation than fiber. So it will have **lower latency**, leading to a lower propagation delay.

Optical fiber

such as Passive Optical networking which can achieve 1Gbps or greater. Efficient and suitable for long distance and high speed. Fiber has **more bandwidth** than copper.

Fiber is generally the best option for data rate and network efficiency.

Bandwidth of fiber optics

The relationship between bandwidth and data rate (symbol rate) can be generalized using **Nyquist's sampling theorem**. It states that in a noiseless channel with bandwidth B, the data rate R can be calculated as twice the channel's bandwidth.

$$R[\text{bit/s}] = B[\text{Hz}]$$

For plastic optical fiber, bandwidth is dependent on the **modal dispersion(temporal spreading)** of the light signal in the fiber. The **multiplex paths in multimode** result in temporal spreading of data, limiting the possible maximum baud rate. The degree of modal dispersion is affected by wavelength of light used, the light launching conditions, the **fiber's refractive index**, and the **run length of the fiber**.

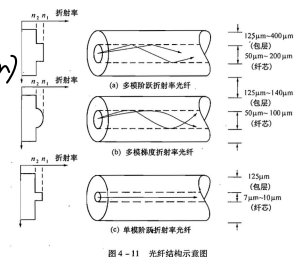
Pulse Dispersion = $k \cdot n_1 \cdot \Delta n \cdot L / c$

Handwritten notes:
fiber's refractive index → multiplex paths in multimode
run (length of fiber)
temporal spreading (modal dispersion)
bandwidth

By controlling these parameters, this dispersion can be reduced to achieve a particular bandwidth or data rate. **Graded-index** POF (GI-POF) limits the number of modes transmitted better than conventional **step-index** POF (SI-POF) does. It has the effect of continuously guiding the light towards the center of the fiber core. This reduces modal dispersion and consequently increases the bandwidth, which is almost double that of SI-POF.

Handwritten note: Graded index fiber → continuously guide the light towards the center of the fiber core
↳ limit the number of modes transmitted → reduces modal dispersion → bandwidth

Cable Type	Maximum Cable Segment	Speed	Bandwidth	Distance
Fiber optic cable	2km - 10km	10/100/1000 Mbps, 10/100/400/800 Gbps	Hundreds of Gigahertz (GHz)	Up to 80km
Twisted pair cable	100m	Up to 10 Gbps Cat5: 100Mbps Cat5e: 100Mbps Cat6: 10Gbps Cat6e: 10Gbps	Cat5: 100MHz Cat5e: 100MHz - 250MHz Cat6: 250MHz - 500MHz Cat6e: 500MHz - 1000MHz	Up to 100m
Coaxial cable	200m - 500m	10Gbps 10Mbps - 100 Mbps	RG6: 1GHz - 2.3GHz RG11: 1GHz - 3GHz	Up to 500m



Channon-Hartley theorem $C = B \log_2(1 + S/N)$

~~Ex. Explain what limits the maximum amount of information that can be transited over a single mode fibre optical communication link and the physical reason for this limit.~~

Frequency bandwidth of the fibre, due to **fibre doping** and **core size**, which will affect the modal dispersion.

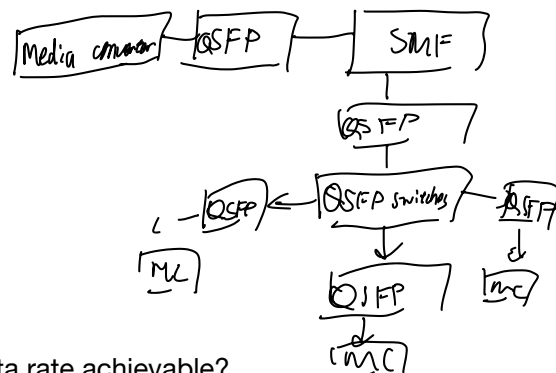
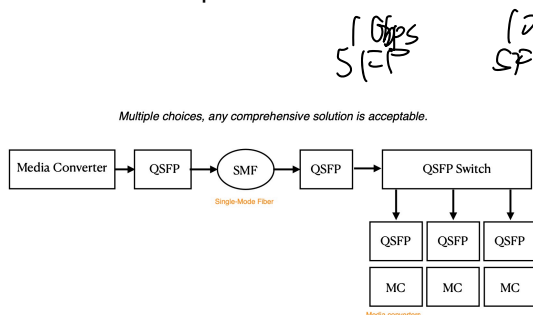
Additionally, **SNR** plays a fundamental role in information capacity, however Stimulated Brillouin scattering (**SBS, or says Electrostriction**) will limit the maximum SNR achievable due to increased optical loss at high pulse power (non-linear Shannon limit).

Ex: Explain the issues that might lead to transmission errors in both single mode and multi-mode optical fibers.

For multimode, due to different modes it has **modal dispersion**, **inter-symbol interference** and they will limit the data rate achievable.

For single-mode, **bend** or **brakes** in the optical fibre will increase losses or lead to back reflection that may lead to increase error signal errors.

Ex. High-capacity low costs interconnects that support up to 100 Gbps are required for a data centre network with fibre spans that are a maximum of 100m in length.



Optical carrier

Ex: why an optical carrier is important for the high data rate achievable?

Baud rate cannot be higher than carrier frequency. Optical carrier is >100THz, therefore high potential for information transmission.

$$\frac{c}{\pi}$$

